The stabiliser of the bulk

What the stabiliser of UM-3 is, why the field equations of the author's metric require it, what it does and what it does not do, and why UM-3 needs it: a complete account for a newcomer, with its provenance

The stabiliser: found and named by Claude (Anthropic), 2026-10-24

This account and its checks: Claude (Anthropic), 2026-10-25

Underlying theory, metric and Lovelock weights: Patrick L. Nash

Repository once-ere/Dirac_claude, folder universe_model_3

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1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \ldots, x_7) :\ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The author's metric depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one fold, at $z = \pi/2$. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \operatorname{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln \sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\operatorname{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the author's metric is

$$g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4} b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2 \tag{FD-4}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8 X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8 X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$, which multiplies ordinary space and the extra times, equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\, g^{ac}\, \partial_\sigma g_{cb}: \qquad K^1{}_1 = K^2{}_2 = K^3{}_3 = K^5{}_5 = K^6{}_6 = K^7{}_7 = -H\,\mathrm{sgn}(\sigma), \qquad K^4{}_4 = 0 \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface $\sigma = \text{const}$ of the hidden direction, $K^a{}_b = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along ordinary space and the extra times, 0 along the time. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

it jumps by $-2H$ along ordinary space and along the extra times. The Einstein-Lovelock equations are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_{\mathrm{D}}(\sigma)$ into them: the metric solves them only if the fold carries a thin source $\mathcal{S}^a{}_b$, an energy-momentum tensor concentrated on $\sigma = 0$ (the junction conditions). At the author's vacuum ($a_4' = H$, b_4 constant) its frame components along (ordinary space, the time, the extra times) are $\kappa\mathcal{S} = (-10H, -6H, -10H)$, at the static late-time state (a_4 and b_4 constant) $\kappa\mathcal{S} = (-61H/8, -31H/4, -61H/8)$, with $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0x_0} = \cot^2(6Hx_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2\Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \dots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-3 the Standard Model (the Higgs doublet with the ad hoc vacuum, the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$, three generations of quarks and leptons and three right-handed neutrinos), the fold fluid and the induced Einstein-Hilbert term of the fold live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \dots, x_4$ (no field depends on the extra times). Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \dots, x_4$. dirac16complex and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the bound states χ of dirac16complex are the part of it that the fold holds, and the portals couple dirac16complex to the Standard Model at $\sigma = 0$. In UM-3 files the word *fold* means this surface and nothing else: a number of equal energy levels is written as a number of degenerate states, and a growth by powers of Euler's number as a logarithm.

Examples of folds that are meaningful for UM-3.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the Standard Model, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. This is why the period $\pi/(6H)$ of x_0 produces no Kaluza-Klein tower with the spacing $12H$; the x_0 mass of the bound states is their binding to the fold.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property 3). In UM-3 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \dots, x_7$, and the crease is the fold.
4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_{\mathrm{D}}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- δ_D: The Dirac delta function: $\int \mathrm{d}\sigma\, \delta_\mathrm{D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every function of UM-3 that depends on σ through $|\sigma|$ is folded at $\sigma = 0$: the metric functions of FD-4 and the profiles of the bound states of dirac16complex (example 7). 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_\mathrm{RS}^2 + \mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_\mathrm{RS} = \frac{6k_\mathrm{RS}}{\kappa_5}, \qquad \Lambda_\mathrm{RS} = -\frac{6k_\mathrm{RS}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y_RS: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-3). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_RS: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp of the brane is $\mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}$ (FD-8, an example, not UM-3). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_RS: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_\mathrm{RS} = 6k_\mathrm{RS}/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, $\kappa_5 = 8\pi G_5$ (FD-8, an example, not UM-3). Unit: GeV^{-3}. Status: STANDARD.
- Λ_RS: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_\mathrm{RS} = -6k_\mathrm{RS}^2/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^5. Status: STANDARD.

has a fold at $y_\mathrm{RS} = 0$ with the properties 1 to 3 (its coordinate y_RS is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_RS (Israel's junction condition) and in the bulk the vacuum energy Λ_RS. The fold of UM-3 is of the same kind; it differs in that the time x_4 is not warped, three extra times are warped together with ordinary space, and the Lovelock terms add to the junction conditions. 7. *A particle bound to a thin layer.* A particle held by an attractive thin layer has a folded wave function: it falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-3 the particles χ of dark matter are the states of dirac16complex bound to the fold; at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega_\mathrm{b}x_4}$)

$$\begin{aligned} &\chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+) \quad (\sigma > 0), \qquad \chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-) \quad (\sigma < 0), \\ &\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R}\big)\Big)\chi(0^-), \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2} \end{aligned} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

where μ_L, μ_R are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the Standard-Model fields on the fold) and $q_\mathrm{b} = -m_\Psi(\mu_\mathrm{L} + \mu_\mathrm{R})\tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$ for $\mu_\mathrm{R} = 0$). 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the bound states of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma)$ (FD-10): it maps the bound states of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of the mean field of a gas of the particles χ: the four bound states of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so the mean field of the gas cannot mix the two signs of the indefinite charge. R_2 maps the bound states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); the gauge fields $E_8 \times E_8$ live on them while gravity fills the bulk. This is the arrangement of UM-3, where the Standard Model lives on the fold and gravity and dirac16complex fill the bulk.

What is not a fold.

- *The tips* $z = 0$ and $z = \pi \pmod{\pi}$: the mirror exchanges them within a period, but there the warp vanishes, $g_{x_0 x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma = 0$, but without a kink and without a thin source.

2 How to read this document

2.1 Who this document is for

It is written for a reader who is clever but new to the subject: someone who knows what a coordinate, a derivative, an energy and a pressure are, but not general relativity, warped geometries or the universe model UM-3. Every term is explained where it is first used, every equation that the document uses is written out in it, and every symbol of every display equation is listed and defined right after the display. Nothing has to be looked up anywhere else.

2.2 The status labels

Every statement carries one of the labels **AUTHOR** (from the author, Patrick L. Nash: his notebooks `Dirac_Einstein-Lovelock_4+4_Nash.nb` and `4+4.nb` and his order of 2026-10-24), **STANDARD** (textbook physics or a published measurement, named in the provenance), **ASSUMED** (a choice made in UM-3 without derivation), **DERIVED** (computed and verified in UM-3; the check that verifies it is named, e.g. ST1), **ESTIMATE** (an order-of-magnitude evaluation) and **OPEN** (not settled).

2.3 Every equation is followed by its symbols

A display equation carries a tag. SB-n is an equation of this document; FD-n belongs to the definition of the fold at the beginning; FE-n (a field equation) and IN-n (a term of the action) are equations of UM-3 that are written identically, with the same tag, in every UM-3 document that uses them. After every display comes the list "Symbols in ..." of every symbol in it, with its meaning, its unit, its value where one exists, and its status. The units are natural units: the reduced Planck constant and the speed of light are 1, so that a time is a length, and energies, masses and inverse lengths are all measured in GeV.

2.4 The checks

The statements labelled DERIVED in this document are verified by eight checks, ST1 to ST8, of a program that computes the field equations exactly (with computer algebra) and compares them with every number written here. The provenance at the end says what each check does and how to run it.

3 The stabiliser in brief

UM-3 is a model universe built on a metric (a rule for measuring distances and durations, explained below) that the author wrote down for an eight-dimensional spacetime. The metric has a fold, the surface defined at the beginning of this document, and around it the bulk: everything that is not on the fold. Gravity in UM-3 obeys the Einstein-Lovelock field equations, which say how the curvature of spacetime is fixed by the energy that it contains.

The finding. When the universe is old and expands slowly, as ours does, the bulk of the author's metric is very nearly static: nothing in it changes with time. A static bulk of this metric is **not** a solution of the field equations with empty space. The equations demand a definite energy content: a negative energy density, and a negative pressure five times as large, the same in every direction of the bulk except the time (SB-8, IN-7). UM-3 calls whatever supplies it **the stabiliser of the bulk**, because without it the bulk could not stay static. The stabiliser has no free parameter: its energy density and its pressure are fixed by the author's constant H and the gravitational coupling κ.

Who found it. The requirement and the stabiliser are not part of the author's specification. They were found, introduced and named by Claude, an AI model made by Anthropic, working in Claude Code sessions configured for the model claude-opus-5-5, on 2026-10-24 while it derived the complete field equations of UM-3. They are checked here in eight checks (ST1 to ST8).

The five points, in brief.

1. *The stabiliser couples only to gravity.* This is how UM-3 defines it: a medium of the bulk that enters the equations only through its energy and its pressure. On the static bulk this is consistent: the stabiliser is conserved by itself (ST4). It is **not** exact once the universe expands. The bulk source that the equations then require releases energy as ordinary space grows, and this energy flows along the hidden direction into the fold (SB-12 to SB-14, FE-65, ST6): the stabiliser and the matter of the fold exchange energy. DERIVED.
2. *It is absent in the primordial era, whose bulk is the author's vacuum (FE-31).* True: the author's vacuum, in which ordinary space expands at the rate H while b_4 is constant, solves the field equations of the bulk with no source at all (ST2). But a medium that couples only to gravity and keeps its pressure at five times its energy density cannot appear later out of nothing (SB-11, ST6). Points 1 and 2 together therefore require that the stabiliser is created through the exchange of energy with the fold, or that its pressure has not always been five times its energy density. UM-3 does not say which. DERIVED (the facts), OPEN (the mechanism).
3. *Its existence and value are DERIVED from the requirement of a static bulk.* The value is DERIVED (ST1). The existence of a bulk source is DERIVED as well, and more generally than for an exactly static bulk: without a source in the bulk the field equations allow no instant at which both ordinary space and the extra times change slowly (FE-66, ST8). That this source is a separate medium that couples only to gravity is ASSUMED, and its nature is OPEN; the only simple field that reproduces it is a field with negative kinetic energy, a ghost (SB-10, ST5).
4. *The static bulk* is explained, with five examples, in the part "The static bulk".
5. *Warp and warp factor* are defined, with four examples, in the part "Warp and warp factor".

Its importance. Without a bulk source the bulk of the author's metric must change at a rate comparable to H (ST3, ST8). A slowly expanding late universe would then require H itself to be as small as today's expansion rate, and then the couplings of the Standard Model would drift more than a million times faster than measurements allow (ST7). The energy density of the stabiliser is the energy density of the bulk at which ordinary space stands still, and the expansion of the universe measures the excess of the energy density of the bulk over it (FE-66). The stabiliser thereby makes a late universe like ours possible in UM-3; together with the static bulk it fixes the source of the fold at late times, and in an expanding universe it powers the fold. Its negative energy density makes it the most exotic ingredient of the model.

4 The language

4.1 Coordinates, distances and the metric

A point of space and time is labelled by numbers, its coordinates. Ordinary space needs three (x_1, x_2, x_3) and time one (x_4). The spacetime of UM-3 has eight: x_0, a hidden direction that we do not see, x_1, x_2, x_3 ordinary space, x_4 the time, and x_5, x_6, x_7 three extra directions that behave like time (the extra times). The metric is the rule that turns a small step of the coordinates into the squared length (or the squared duration) of the step. A familiar example is the surface of a globe of radius R_g, on which a point is labelled by its angle θ_g from the north pole and its longitude ϕ_g:

$$\mathrm{d}\ell_\mathrm{g}^2 = R_\mathrm{g}^2\,\mathrm{d}\theta_\mathrm{g}^2 + R_\mathrm{g}^2 \sin^2\theta_\mathrm{g}\,\mathrm{d}\phi_\mathrm{g}^2 \qquad \text{(SB-1)}$$

Symbols in (SB-1).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- ℓ_g: The length measured along the surface of a globe; $\mathrm{d}\ell_\mathrm{g}$ is the length of a small step on it (an example of a metric, not UM-3). Unit: length. Status: STANDARD.
- R_g: The radius of a globe (an example, not UM-3). Unit: length. Status: STANDARD.
- θ_g: The angle of a point of a globe from its north pole: 0 at the north pole, $\pi/2$ on the equator, π at the south pole (an example, not UM-3). Unit: radian. Status: STANDARD.
- ϕ_g: The longitude of a point of a globe (an example, not UM-3). Unit: radian. Status: STANDARD.

A step of the longitude by $\mathrm{d}\phi_\mathrm{g}$ is long on the equator ($\sin\theta_\mathrm{g} = 1$) and short near the poles: the factor $\sin^2\theta_\mathrm{g}$ in front of $\mathrm{d}\phi_\mathrm{g}^2$ says how large one unit of longitude is at each latitude. Status: STANDARD.

The metric of the eight-dimensional spacetime is written $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$: the indices M and N run over the eight coordinates $X = \{x_0, \ldots, x_7\}$, a repeated index is summed, and the numbers g_{MN} are the components of the metric.

In spacetime a step can be space-like (a positive squared interval: a distance) or time-like (a negative one: minus the square of a duration). UM-3 counts x_0, x_1, x_2, x_3 as space-like and x_4, x_5, x_6, x_7 as time-like (the signature $+,+,+,+,-,-,-,-$). The author's metric is written in FD-4 of the definition of the fold with the proper distance σ from the fold: $\mathrm{d}\sigma^2$ for the hidden direction, a factor that depends on σ in front of ordinary space and of the extra times, and $-\mathrm{d}x_4^2$ for the time. The two functions $a_4(x_4)$ and $b_4(x_4)$ of the time are the author's: e^{a_4} sets the size of ordinary space and $\mathrm{e}^{-a_4}b_4$ the size of the extra times on the fold. Status: AUTHOR (the metric), DERIVED (its form in σ, FD-3 and FD-4).

4.2 Warp and warp factor

Definition. A metric is *warped* along a direction when the size of other directions depends on the position along it. In a metric of the form $\mathrm{d}\sigma^2 + \mathcal{W}(\sigma)\,\mathrm{d}y^2 + \ldots$ the function $\mathcal{W}(\sigma)$ that multiplies the squared steps $\mathrm{d}y^2$ of the other directions is the **warp factor**. The length of a fixed coordinate step $\mathrm{d}y$ is $\sqrt{\mathcal{W}}\,\mathrm{d}y$, so lengths along those directions scale with $\sqrt{\mathcal{W}}$. The **warp** is the dependence itself, measured by the rate at which $\ln\mathcal{W}$ changes along σ. In UM-3:

$$\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|} = (\sin^2 z)^{1/6}, \qquad \frac{\mathrm{d}\ln\mathcal{W}}{\mathrm{d}\sigma} = -2H\,\mathrm{sgn}(\sigma) \tag{SB-2}$$

Symbols in (SB-2).

- $\mathcal{W}$: The warp factor: in a warped metric, the function of the position along one direction that multiplies the squared steps of other directions; in UM-3 the function of σ that multiplies ordinary space and the extra times, $\mathcal{W} = \mathrm{e}^{-2H|\sigma|} = (\sin^2 z)^{1/6}$, equal to 1 on the fold. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The warp factor multiplies ordinary space (together with e^{2a_4}) and the extra times (together with $\mathrm{e}^{-2a_4}b_4^2$), but not the time x_4 and not σ itself. It is largest, 1, on the fold and falls on both sides. A coordinate step of ordinary space at the depth $|\sigma|$ in the bulk is $\mathrm{e}^{-H|\sigma|}$ times as long as the same coordinate step on the fold: the deeper in the bulk, the smaller ordinary space and the extra times become. The warp factor is the same on both sides of the fold, and the warp, its rate of change, changes its sign on the fold: this is the kink of the fold (property 3 of the definition). Status: AUTHOR (the metric), DERIVED (the form in σ).

Examples of warps.

- *A globe* (SB-1): the longitude is warped along the angle from the pole, with the warp factor $\sin^2\theta_\mathrm{g}$.
- *The fold of UM-3* (SB-2): ordinary space and the extra times are warped along the hidden direction.
- *The brane of Randall and Sundrum* (FD-8, example 6 of the definition of the fold): the four directions of our world are warped by $\mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}$ along a fifth direction. Unlike in UM-3, the time is warped as well.
- *Time near the Earth.* Near a mass the time is warped. In the weak gravity of the Earth the factor in front of $-\mathrm{d}x_4^2$ is $1 + 2\Phi_\mathrm{N}$, where Φ_N is the Newtonian gravitational potential (negative, and larger in size closer to the Earth). A clock lower down therefore runs slower. Pound and Rebka measured this in 1960 over the 22.5 m height of a tower, and the clocks of the GPS satellites are corrected for it. In UM-3 the time is not warped along the hidden direction: a clock deep in the bulk runs at the same rate as a clock on the fold.

4.3 The bulk

Definition. The bulk is all of the eight-dimensional spacetime except the fold: the points with $\sigma \neq 0$. It extends on both sides of the fold to the tips, which lie at an infinite proper distance. Gravity and dirac16complex live in the

bulk as well as on the fold; dirac16complex is the author's field of matter of spin one half in eight dimensions, a spinor with sixteen complex components. The Standard Model lives only on the fold. Status: ASSUMED (where the fields live), AUTHOR (the metric and dirac16complex).

4.4 Energy, pressure and the energy-momentum tensor

Matter and fields carry energy and momentum. Their densities and flows are collected in the energy-momentum tensor, an eight-by-eight array whose entries are measured along the eight directions of an orthonormal frame: the frame direction (0) along σ, $(1), (2), (3)$ along ordinary space, (4) along the time and $(5), (6), (7)$ along the extra times. For a medium at rest the entry along the time is minus its energy density, the entries along the other directions are its pressures, and an entry that mixes the time with another direction is a flow of energy along that direction. A medium at rest with the same pressure p_{med} in every direction but the time has

$$\mathcal{T}_{\mathrm{med}}{}^{a}{}_{b} = \mathrm{diag}\big(p_{\mathrm{med}}, p_{\mathrm{med}}, p_{\mathrm{med}}, p_{\mathrm{med}}, -\rho_{\mathrm{med}}, p_{\mathrm{med}}, p_{\mathrm{med}}, p_{\mathrm{med}}\big) \tag{SB-3}$$

Symbols in (SB-3).

- $\mathcal{T}_{\mathrm{med}}{}^{a}{}_{b}$: The energy-momentum tensor of one medium (frame components): for a medium at rest the entry along the time is minus its energy density, the entries along the other directions are its pressures, and the entries that mix the time with another direction are its flows of energy along that direction. Unit: GeV^8 (energy per volume in eight dimensions). Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, $(1), (2), (3)$ along ordinary space, (4) along the time, $(5), (6), (7)$ along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_{med}: The pressure of a medium at rest, the same along every direction but the time (for a medium of the bulk: along σ, ordinary space and the extra times). Unit: GeV^8. Status: STANDARD.
- ρ_{med}: The energy density (energy per volume) of a medium at rest. Unit: GeV^8. Status: STANDARD.

Three examples in the four dimensions of our world: dust (pressureless matter, $p_{\mathrm{med}} = 0$), light ($p_{\mathrm{med}} = \rho_{\mathrm{med}}/3$) and the energy of the vacuum ($p_{\mathrm{med}} = -\rho_{\mathrm{med}}$; a vacuum energy pulls instead of pushing). The energy density of every known form of matter and radiation is positive; a medium with a negative energy density is called exotic. Status: STANDARD.

4.5 The field equations of gravity

Einstein's insight was that gravity is the curvature of spacetime, and that the curvature is fixed by the energy-momentum tensor. One side of his field equations is a combination of derivatives of the metric (the Einstein tensor); the other is the energy-momentum tensor times the gravitational coupling. UM-3 uses the generalisation found by Lovelock, which in eight dimensions adds two more combinations of the curvature, each with a weight chosen by the author. The action (the quantity whose variation gives the equations) and its curvature terms are

$$\begin{aligned} S_{\mathrm{EL}} &= \frac{1}{2\kappa} \int \mathrm{d}^8 X \, \sqrt{|g|} \, \Big(\sum_{p=1}^{3} \alpha_{(p)} \, \mathcal{L}_{(p)} - 2\Lambda \Big), \qquad \kappa = 8\pi G_8, \\ \alpha_{(p)} &= w_{(p)} \, H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}, \qquad \Lambda = -10H^2 \end{aligned} \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_\mathrm{N} = 10$ in $0 = -\Lambda_\mathrm{N} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]). Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \end{aligned} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.

and the field equations of the bulk are

$$\sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\big(\mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_\mathrm{stab}{}^M{}_N\big) \qquad (\sigma \neq 0) \tag{FE-1}$$

Symbols in (FE-1).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- ${\mathcal{T}_{\mathrm{stab}}}^{a}{}_{b}$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric requires, found and named by Claude (Anthropic) on 2026-10-24. Unit: GeV^8. Status: DERIVED (the value), ASSUMED (a separate medium).
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The left-hand side is built from the metric alone. The right-hand side is the energy content of the bulk: dirac16complex and the stabiliser. A metric solves the equations only if the right-hand side has exactly the value that the left-hand side computes; a solution with nothing on the right is a *vacuum*. Status: AUTHOR (the weights and Λ; the author writes the pure number 10 in his vacuum equations), STANDARD (Lovelock gravity), ASSUMED (the stabiliser on the right).

The divergence of every Lovelock tensor vanishes identically (the Bianchi identity). The right-hand side must therefore be conserved: the energy and momentum of all sources together cannot be created or destroyed, only moved. A single medium that interacts with nothing but gravity is conserved by itself:

$$\nabla_M \, {\mathcal{T}_{\mathrm{med}}}^{M}{}_{N} = 0 \tag{SB-4}$$

Symbols in (SB-4).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols ${\Gamma^{P}}_{MN}$). Status: STANDARD.
- ${\mathcal{T}_{\mathrm{med}}}^{a}{}_{b}$: The energy-momentum tensor of one medium (frame components): for a medium at rest the entry along the time is minus its energy density, the entries along the other directions are its pressures, and the entries that mix the time with another direction are its flows of energy along that direction. Unit: GeV^8 (energy per volume in eight dimensions). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

(∇ is the covariant derivative, the derivative that takes into account how the directions of the coordinates change from point to point.) Status: STANDARD.

5 The static bulk

5.1 What static means

Definition. A spacetime is *static* when nothing in it changes with the time and nothing flows: its metric does not depend on the time, and no motion that the reversal of time would change takes place in it. In the author's metric the time dependence sits in the two functions $a_4(x_4)$ and $b_4(x_4)$. The bulk is static when both are constant, that is when the rates $\mathsf{A} = a_4'/H$ (the rate at which ordinary space grows) and $\mathsf{U} = b_4'/(Hb_4)$ (the rate at which b_4 changes) both vanish:

$$\begin{aligned} &\mathsf{A} = \frac{a_4'}{H} = 0, \qquad \mathsf{U} = \frac{b_4'}{Hb_4} = 0 : \\ &g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2 \end{aligned} \tag{SB-5}$$

Symbols in (SB-5).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

with constant a_4 and b_4 (a prime is the derivative with respect to the time x_4). UM-3 calls this the *static block*.

Status: DERIVED (the form), ASSUMED (that the late bulk is the static block to first approximation).

5.2 Static is not the same as empty

A static spacetime may be empty or it may need a source, and an empty spacetime may be static or not:

spacetime	static?	empty (a vacuum)?
the static block of UM-3 (SB-5)	yes	no: it needs the stabiliser (SB-8)
the author's vacuum (FE-31)	no: ordinary space grows at the rate H	yes
the space outside a star	yes	yes
Einstein's static universe (SB-6)	yes	no: it needs dust and a cosmological constant

5.3 Examples of static spacetimes and static media

Example 1: the static block of UM-3 (SB-5). It is the late universe to first approximation. Today ordinary space expands at the Hubble rate $H_4^{(0)} = 1.44 \times 10^{-42}$ GeV. UM-3 takes the author's constant H much larger, so the rate $\mathsf{A} = H_4^{(0)}/H$ is tiny and the bulk is static up to corrections of order A. The extra times are frozen ($\mathrm{e}^{-a_4}b_4$ constant, $\mathsf{U} = \mathsf{A}$), so that the couplings of the Standard Model do not change. Status: ASSUMED.

Example 2: the space outside a star or a planet (Schwarzschild's solution of 1916). It is static and empty: a static vacuum. Not every static spacetime needs a source. Status: STANDARD.

Example 3: Einstein's static universe (1917). Einstein wanted a universe that does not change. With dust of the mass density ρ_E and nothing else his equations do not allow it, because gravity pulls the dust together. He added the cosmological constant Λ_E, which pushes apart, and found a static universe, a three-dimensional sphere of radius R_E, but only for one exact balance:

$$\Lambda_\mathrm{E} = 4\pi G_\mathrm{N}\,\rho_\mathrm{E} = \frac{1}{R_\mathrm{E}^2} \tag{SB-6}$$

Symbols in (SB-6).

- Λ_E: The cosmological constant of Einstein's static universe of 1917, in four dimensions (an example, not UM-3). Unit: 1/length^2. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$ with the Planck mass $M_\mathrm{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- ρ_E: The mass density of the dust that fills Einstein's static universe (an example, not UM-3). Unit: mass per volume. Status: STANDARD.
- R_E: The radius of the three-dimensional sphere that is the space of Einstein's static universe (an example, not UM-3). Unit: length. Status: STANDARD.

This is the classic example of a static spacetime that requires a definite source. It is also unstable (Eddington, 1930): a small excess of matter makes it collapse, a small deficit makes it expand for ever. Einstein's Λ_E plays for his universe the part that the stabiliser plays for the bulk of UM-3. Status: STANDARD.

Example 4: the air at rest above the ground. The atmosphere stays at rest only because its pressure falls with the altitude exactly so fast that the difference of pressure carries the weight of the air in between:

$$\frac{\mathrm{d}P_\mathrm{air}}{\mathrm{d}h_\mathrm{alt}} = -\rho_\mathrm{air}\, g_\oplus \tag{SB-7}$$

Symbols in (SB-7).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

- P_{air}: The pressure of the air at the altitude h_{alt} (an example, not UM-3). Unit: Pa. Value: about 1.0×10^5 Pa at the ground. Status: STANDARD.
- h_{alt}: The altitude above the ground (an example, not UM-3). Unit: m. Status: STANDARD.
- ρ_{air}: The mass density of the air at the altitude h_{alt} (an example, not UM-3). Unit: kg/m^3. Value: about 1.2 kg/m^3 at the ground. Status: STANDARD.
- $g_\oplus$: The acceleration of free fall at the surface of the Earth (an example, not UM-3). Unit: m/s^2. Value: 9.81 m/s^2. Status: STANDARD.

Again, the static state needs a definite distribution of pressure; any other distribution sets the air in motion. Status: STANDARD.

Example 5: a drum skin at rest. A stretched skin stays flat and at rest because its tension is the same everywhere, so that the pulls on every small piece of it balance. Pressing it or loosening it locally sets it in motion. Status: STANDARD.

In all these examples except the empty space outside a star, staying static requires a definite content: a definite balance of energy and pressure. The static block of UM-3 is of this kind.

6 The stabiliser

6.1 The requirement of a static bulk

Insert the static block SB-5 into the left-hand side of FE-1 and evaluate it with the author's weights and $\Lambda = -10H^2$. The computation (exact, by computer algebra, ST1) gives, on both sides of the fold and with no entry that mixes two directions,

$$\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{a}{}_{b} + \Lambda\, \delta^a_b = H^2\, \mathrm{diag}\Big(-\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, \frac{1}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16} \Big) \qquad (\mathsf{A} = \mathsf{U} = 0,\ \sigma \neq 0) \qquad \text{(SB-8)}$$

Symbols in (SB-8).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: 1/length^{2p}. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.

- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

in the order (σ, x_1, x_2, x_3, x_4, x_5, x_6, x_7). The static bulk is therefore not a vacuum: the field equations hold only if the bulk contains a source whose energy-momentum tensor is this left-hand side divided by κ. In the language of SB-3 it is a medium at rest, without any flow, with the energy density ρ_s and the same pressure p_s in every direction but the time. UM-3 states it as a term of its action:

$$\mathcal{T}_\mathrm{stab}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, -\rho_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}\big), \qquad \kappa\rho_\mathrm{s} = -\frac{H^2}{16}, \qquad \kappa p_\mathrm{s} = -\frac{5H^2}{16} \tag{IN-7}$$

Symbols in (IN-7).

- $\mathcal{T}_\mathrm{stab}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric requires, found and named by Claude (Anthropic) on 2026-10-24. Unit: GeV^8. Status: DERIVED (the value), ASSUMED (a separate medium).
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_s: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_\mathrm{s} = -5H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).
- ρ_s: The energy density of the stabiliser; with frozen extra times it is the energy density of the bulk at which ordinary space stands still. Unit: GeV^8. Value: $\kappa\rho_\mathrm{s} = -H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The same static block, together with the source on the fold that its kink requires, is the field equation FE-32 of UM-3:

$$\begin{aligned} &\mathsf{A} = \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad \kappa\,\mathcal{T}_\mathrm{bulk}{}^a{}_b = H^2\,\mathrm{diag}\Big(-\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, \frac{1}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}\Big), \\ &\kappa\,\mathcal{S}^{(i)}{}_{(i)} = \kappa\,\mathcal{S}^{(r)}{}_{(r)} = -\frac{61}{8}H, \qquad \kappa\,\mathcal{S}^{(4)}{}_{(4)} = -\frac{31}{4}H \end{aligned} \tag{FE-32}$$

Symbols in (FE-32).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\mathrm{bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_\mathrm{stab}$; $\mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); it is fixed by the jump of the first σ-derivatives of the metric across the fold (the junction conditions; $\kappa\mathcal{S}$ is the jump of the junction potential). Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ a direction x_i of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

Here A_1 and U_1 are the rates at which A and U change, $\mathcal{T}_{\mathrm{bulk}}$ is the energy-momentum tensor of all sources of the bulk together, and $\mathcal{S}$ is the thin source of the fold (property 3 of the definition of the fold). Status: DERIVED (ST1).

6.2 Why it is called the stabiliser, and who found it

It is the energy content without which the bulk of the author's metric cannot stay static: it *stabilises* the bulk in the sense of keeping it static. (Whether the static bulk with the stabiliser is also stable, that is whether small disturbances die away, is a separate question; it is OPEN, see the part on its importance.) The stabiliser is not part of the author's specification. Claude (Anthropic's AI model, working in Claude Code) found the requirement SB-8 on 2026-10-24 while deriving the complete coupled field equations of UM-3, introduced the medium IN-7 to meet it, named it the stabiliser of the bulk, and verified it in the checks of this document. Status: DERIVED (the requirement), ASSUMED (the medium).

6.3 What kind of medium it is

Its energy density and its pressure are negative. $\rho_{\mathrm{s}} = -H^2/(16\kappa)$, and the pressure is five times as large, $p_{\mathrm{s}} = 5\rho_{\mathrm{s}}$, the same along the hidden direction, ordinary space and the extra times. It does not flow.

It violates the energy conditions. The weak energy condition (the energy density is never negative) and the null energy condition (the energy density plus the pressure in any direction is never negative) both fail:

$$\kappa\rho_{\mathrm{s}} = -\frac{H^2}{16} < 0, \qquad \kappa\big(\rho_{\mathrm{s}} + p_{\mathrm{s}}\big) = -\frac{3H^2}{8} < 0 \tag{SB-9}$$

Symbols in (SB-9).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{s}: The energy density of the stabiliser; with frozen extra times it is the energy density of the bulk at which ordinary space stands still. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{s}} = -H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- p_{s}: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_{\mathrm{s}} = -5H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).

Ordinary matter, light and a vacuum energy satisfy both. Status: DERIVED (ST5).

Only a ghost field reproduces it. A field that fills the bulk uniformly and changes only with the time has a kinetic energy K_{f} and a potential energy V_{f} per volume, and its energy density and pressure (the same in every direction but the time) are

$$\rho_{\mathrm{med}} = K_{\mathrm{f}} + V_{\mathrm{f}}, \qquad p_{\mathrm{med}} = K_{\mathrm{f}} - V_{\mathrm{f}}; \qquad p_{\mathrm{med}} = 5\rho_{\mathrm{med}}: \quad V_{\mathrm{f}} = -\frac{2}{3}K_{\mathrm{f}}, \quad \rho_{\mathrm{med}} = \frac{K_{\mathrm{f}}}{3} \tag{SB-10}$$

Symbols in (SB-10).

- ρ_{med}: The energy density (energy per volume) of a medium at rest. Unit: GeV^8. Status: STANDARD.
- K_{f}: The kinetic energy per volume of a field that fills the bulk uniformly and changes only with the time x_4: half the square of its rate of change for an ordinary field (never negative), minus that for a field whose kinetic term has the opposite sign (a ghost). Unit: GeV^8. Status: STANDARD.
- V_{f}: The potential energy per volume of a field that fills the bulk uniformly and changes only with the time x_4. Unit: GeV^8. Status: STANDARD.
- p_{med}: The pressure of a medium at rest, the same along every direction but the time (for a medium of the bulk: along σ, ordinary space and the extra times). Unit: GeV^8. Status: STANDARD.

An ordinary field has $K_{\mathrm{f}} \geq 0$ and so, with a pressure five times its energy density, a positive energy density. Only a field whose kinetic energy is negative, a so-called ghost, with $K_{\mathrm{f}} = 3\rho_{\mathrm{s}}$ and $V_{\mathrm{f}} = -2\rho_{\mathrm{s}}$, reproduces the stabiliser; on the static bulk it rolls at constant speed on a flat potential. A field that depends on σ instead of the time cannot do it at all, because its pressure along σ differs from its pressure along the other directions unless it is constant, and a constant field is a vacuum energy with $p = -\rho$. Status: DERIVED (ST5).

If it were a simple fluid, its sound would be faster than light. For a fluid whose pressure is a fixed multiple of its energy density, small compressions travel at the speed of sound c_{s} with $c_{\mathrm{s}}^2 = \mathrm{d}p/\mathrm{d}\rho$; for the stabiliser this is 5, so its sound would travel $\sqrt{5} \approx 2.24$ times faster than light. The ghost field instead propagates at the speed of light, but its quanta carry negative energy. A general theorem (Dubovsky, Grégoire, Nicolis and Rattazzi, 2006) states that, under broad conditions, a medium that violates the null energy condition has either signals faster than light or unstable excitations. Status: STANDARD (the theorem and the fluid formula), DERIVED (their application).

It is conserved by itself on the static bulk. The covariant divergence SB-4 of the constant tensor IN-7 vanishes on the static block, in every component (ST4). The reason: its pressure along the hidden direction equals its pressure along every warped direction, and the time, along which its entry differs, is not warped. Unlike the air of SB-7, the stabiliser therefore needs no change of pressure along the hidden direction: because the time is not warped there is no gravitational pull along σ, and a medium that couples only to gravity can sit in the static bulk unchanged. Status: DERIVED (ST4).

6.4 How it would evolve on its own

A medium that interacts only through gravity obeys SB-4. For a medium that fills the bulk uniformly, with the energy density $\rho_{\mathrm{med}}(x_4)$ and the pressure $p_{\mathrm{med}}(x_4)$ in every warped direction, in the author's metric with arbitrary $a_4(x_4)$ and $b_4(x_4)$, the time component of SB-4 reads (ST6)

$$\rho'_{\mathrm{med}} = -3\,\frac{b'_4}{b_4}\,\big(\rho_{\mathrm{med}} + p_{\mathrm{med}}\big); \qquad p_{\mathrm{med}} = 5\rho_{\mathrm{med}}: \quad \rho_{\mathrm{med}}\, b_4^{18} = \text{constant} \tag{SB-11}$$

Symbols in (SB-11).

- ρ_{med}: The energy density (energy per volume) of a medium at rest. Unit: GeV^8. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a'_4 = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- p_{med}: The pressure of a medium at rest, the same along every direction but the time (for a medium of the bulk: along σ, ordinary space and the extra times). Unit: GeV^8. Status: STANDARD.

The growth of ordinary space drops out, because ordinary space grows with e^{a_4} and the extra times with $\mathrm{e}^{-a_4} b_4$, so that the volume of all six warped directions together grows with b_4^3. Two consequences follow. A stabiliser that is absent ($\rho = 0$) stays absent. And with frozen extra times (b_4 proportional to e^{a_4}) a stabiliser on its own would be diluted as $\mathrm{e}^{-18 a_4}$ while the universe expands. Status: DERIVED (ST6).

6.5 What the field equations require when the universe expands

The field equations do not ask for such a medium once ordinary space grows. With frozen extra times, the condition of the late universe, they require in the bulk and on the fold (FE-33 of UM-3):

$$
\begin{aligned}
&\mathsf{U}=\mathsf{A},\quad \mathsf{U}_1=\mathsf{A}_1\ \ \big(\mathrm{e}^{-a_4}b_4=\text{constant},\ \mathcal{Z}=1\big):\\
&\frac{\kappa\rho_{\text{bulk}}}{H^2}=\frac{3\mathsf{A}^2-1}{16},\qquad \frac{\kappa\,{\mathcal{T}_{\text{bulk}}}^{(0)}{}_{(0)}}{H^2}=-\frac{3\mathsf{A}^4+3\mathsf{A}^2\mathsf{A}_1+18\mathsf{A}^2+3\mathsf{A}_1+5}{16},\\
&\frac{\kappa\,{\mathcal{T}_{\text{bulk}}}^{(i)}{}_{(i)}}{H^2}=-\frac{9\mathsf{A}^2+2\mathsf{A}_1+5}{16},\qquad \frac{\kappa\,{\mathcal{T}_{\text{bulk}}}^{(r)}{}_{(r)}}{H^2}=-\frac{3\mathsf{A}^4+3\mathsf{A}^2\mathsf{A}_1+24\mathsf{A}^2+3\mathsf{A}_1+5}{16},\\
&\frac{\kappa\,{\mathcal{T}_{\text{bulk}}}^{(0)}{}_{(4)}}{H^2}=\operatorname{sgn}(\sigma)\,\frac{3\mathsf{A}(1+\mathsf{A}^2)}{16},\qquad \frac{\kappa\,{\mathcal{S}}^{(i)}{}_{(i)}}{H}=-\frac{57\mathsf{A}^2+40\mathsf{A}_1+183}{24},\\
&\frac{\kappa\,{\mathcal{S}}^{(4)}{}_{(4)}}{H}=-\frac{10\mathsf{A}^2+31}{4},\qquad \frac{\kappa\,{\mathcal{S}}^{(r)}{}_{(r)}}{H}=-\frac{2\mathsf{A}^4+2\mathsf{A}^2\mathsf{A}_1+37\mathsf{A}^2+20\mathsf{A}_1+61}{8}
\end{aligned}
\tag{FE-33}
$$

Symbols in (FE-33).

- U: The rate of change of b_4 in units of H, $\mathsf{U}=b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U}=\mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A}=a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1=\mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1=\mathsf{A}'/H=a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4>0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\text{bulk}}=-{\mathcal{T}_{\text{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda=-10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathcal{T}_{\text{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi+\mathcal{T}_{\text{stab}}$; ${\mathcal{T}_{\text{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i=1,2,3$), which also labels the neutrino mass states ν_1,ν_2,ν_3; in a frame component such as ${\mathcal{S}}^{(i)}{}_{(i)}$ a direction x_i of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r\in\{5,6,7\}$ (x_5,x_6,x_7); not summed (the three components are equal). Status: STANDARD.
- sgn: The sign function: $\operatorname{sgn}(y)=+1$ for $y>0$, -1 for $y<0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\operatorname{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); it is fixed by the jump of the first σ-derivatives of the metric across the fold (the junction conditions; $\kappa\mathcal{S}$ is the jump of the junction potential). Unit: GeV^7. Status: DERIVED.

The first line is the condition of frozen extra times: the physical size $\mathrm{e}^{-a_4}b_4$ of the extra times stays constant, so b_4 grows with e^{a_4}, and the factor $\mathcal{Z}$ by which the extra times rescale every coupling of the fold stays 1. The next lines are the bulk source and the source of the fold that the field equations then require. For $\mathsf{A}\to0$ they reduce to the static block FE-32. With frozen extra times energy flows along the hidden direction: the entry $\kappa{\mathcal{T}_{\text{bulk}}}^{(0)}{}_{(4)}$ of FE-33 is this flow, equal in size on the two sides of the fold and opposite in sign, so it jumps across the fold by

$$
\kappa\Big({\mathcal{T}_{\text{bulk}}}^{(0)}{}_{(4)}\big|_{\sigma=0^+}-{\mathcal{T}_{\text{bulk}}}^{(0)}{}_{(4)}\big|_{\sigma=0^-}\Big)=\frac{3}{8}\,\mathsf{A}\big(1+\mathsf{A}^2\big)H^2\qquad(\mathsf{U}=\mathsf{A})
\tag{SB-12}
$$

Symbols in (SB-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.

The flow is one of the two components of the bulk equations that contain no second time derivative of the metric; FE-66 below gives it for any $a_4(x_4)$ and $b_4(x_4)$, and it vanishes while b_4 is constant, as in the author's vacuum. The energy current is minus the entry ${\mathcal{T}^{(0)}}_{(4)}$; it points towards the fold on both sides, so in an expanding universe ($\mathsf{A} > 0$) energy flows out of the bulk source into the fold, at the rate SB-12. Status: DERIVED (ST6, ST8).

Where the energy comes from. With frozen extra times only ordinary space grows, and the volume of the bulk grows at the rate $3a_4'$. The bulk source of FE-33 has, for constant A, an energy density plus pressure along ordinary space that is negative:

$$\begin{aligned} &\kappa\big(\rho_{\mathrm{bulk}} + {\mathcal{T}_{\mathrm{bulk}}}^{(1)}{}_{(1)}\big) = -\frac{3}{8}\big(1+\mathsf{A}^2\big)H^2 \qquad (\mathsf{U} = \mathsf{A},\ \mathsf{A}_1 = 0), \\ &\int_{-\infty}^{\infty} \mathrm{d}\sigma\ \mathrm{e}^{-6H|\sigma|}\ \big(-3a_4'\big)\,\kappa\big(\rho_{\mathrm{bulk}} + {\mathcal{T}_{\mathrm{bulk}}}^{(1)}{}_{(1)}\big) = \frac{3}{8}\,\mathsf{A}\big(1+\mathsf{A}^2\big)H^2 \end{aligned} \tag{SB-13}$$

Symbols in (SB-13).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

Left to itself, a medium of the bulk changes its energy density at the rate $-3a_4'(\rho + p)$ while only ordinary space grows (ρ its energy density, p its pressure along ordinary space). This is the time component of SB-4 for a medium that does not flow, as in SB-11: with frozen extra times only the three directions of ordinary space grow, each at the rate a_4', so only the pressure along them enters. Ordinary matter, with $\rho + p > 0$, is diluted. The required source,

with $\rho + p < 0$, would instead see its negative energy density rise towards zero; since the field equations keep its energy density fixed, it must give off the energy $-3a_4'(\rho_{\mathrm{bulk}} + \mathcal{T}_{\mathrm{bulk}}{}^{(1)}{}_{(1)})$ per volume and time. Integrated over the hidden direction, with the volume factor $\mathrm{e}^{-6H|\sigma|}$ of the six warped directions, this is exactly the jump of the flow SB-12: all the energy that the bulk gives off arrives at the fold. Status: DERIVED (ST6).

Where the energy goes. The source of the fold at late times is supplied by the fold fluid, a medium of the fold that UM-3 assumes. At the static block it is a tension λ_{F} (a vacuum energy of the seven-dimensional fold) plus a pressureless part ρ_{d} (FE-65 of UM-3):

$$\rho_{\mathrm{ff}} = \lambda_{\mathrm{F}} + \rho_{\mathrm{d}}, \qquad p_{\mathrm{ff}} = q_{\mathrm{ff}} = -\lambda_{\mathrm{F}}, \qquad \lambda_{\mathrm{F}} = \frac{61HV_*}{8\kappa}, \qquad \rho_{\mathrm{d}} = \frac{HV_*}{8\kappa}, \qquad 3a_4'\,\rho_{\mathrm{d}} = \frac{3}{8}\,\mathsf{A}\,\frac{H^2V_*}{\kappa} \tag{FE-65}$$

Symbols in (FE-65).

- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- λ_{F}: The tension of the fold fluid: the part of the fold fluid whose pressures along ordinary space and along the extra times are minus its energy density (a vacuum energy of the seven-dimensional fold), per unit volume of ordinary space of the fold. Unit: GeV^4. Value: $61HV_*/(8\kappa)$ at the static block. Status: DERIVED.
- ρ_{d}: The pressureless part of the fold fluid: the energy density of the fold fluid minus its tension, per unit volume of ordinary space of the fold. Unit: GeV^4. Value: $HV_*/(8\kappa)$ at the static block. Status: DERIVED.
- p_{ff}: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- q_{ff}: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.

(the energy density and the pressures per unit volume of ordinary space on the fold, after the integration over the extra times of physical volume V_*). A tension does not dilute when the fold grows, but the pressureless part does, and to keep its density it needs the energy $3a_4'\rho_{\mathrm{d}}$ per volume and time: exactly the part of the jump of SB-12 that is linear in A, times V_*. The rest of the jump, of order A^3, is taken up by the part of the source of the fold of FE-33 that grows with A^2. For frozen extra times and any $a_4(x_4)$ the source of the fold of FE-33 takes up exactly the energy that arrives from the bulk:

$$-\big(\kappa\,\mathcal{S}^{(4)}{}_{(4)}\big)' + 3a_4'\big(\kappa\,\mathcal{S}^{(i)}{}_{(i)} - \kappa\,\mathcal{S}^{(4)}{}_{(4)}\big) = \kappa\Big(\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_{(4)}\big|_{\sigma=0^+} - \mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_{(4)}\big|_{\sigma=0^-}\Big) \qquad (\mathsf{U} = \mathsf{A}) \tag{SB-14}$$

Symbols in (SB-14).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); it is fixed by the jump of the first σ-derivatives of the metric across the fold (the junction conditions; $\kappa\mathcal{S}$ is the jump of the junction potential). Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ a direction x_i of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.

($-\mathcal{S}^{(4)}{}_{(4)}$ is the energy density of the source of the fold and $\mathcal{S}^{(i)}{}_{(i)}$ its pressure along ordinary space). This is the second line of the conservation law FE-11 of UM-3, which books the flow out of the bulk on the fold:

$$\begin{aligned} &\nabla_M \big(\mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_{\mathrm{stab}}{}^M{}_N\big) = 0 \qquad (\sigma \neq 0), \\ &\hat{\nabla}_a\, \mathcal{S}^a{}_b = -\Big(\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^+} - \mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b\big|_{\sigma=0^-}\Big) \end{aligned} \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi/\delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- $\mathcal{T}_{\mathrm{stab}}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric requires, found and named by Claude (Anthropic) on 2026-10-24. Unit: GeV^8. Status: DERIVED (the value), ASSUMED (a separate medium).
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat{\nabla}_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); it is fixed by the jump of the first σ-derivatives of the metric across the fold (the junction conditions; $\kappa\mathcal{S}$ is the jump of the junction potential). Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

In an expanding universe the bulk and the fold therefore exchange energy, and only their total is conserved. Status: DERIVED (ST6), STANDARD (the conservation law at a thin source), ASSUMED (the fold fluid).

7 The five points, verified

[1] The stabiliser couples only to gravity. By its definition it has no term in the action of UM-3 other than its own: it enters the field equations only through its energy-momentum tensor in FE-1, and it exerts no force on

dirac16complex or on the Standard Model except through the curvature that it causes. On the static bulk this is consistent: the stabiliser is conserved by itself (ST4). In an expanding universe the bulk source that the equations require releases energy (SB-13) that flows into the fold (SB-12) and is taken up there (FE-65, SB-14). If that source is the stabiliser, the stabiliser exchanges energy with the matter of the fold, which is an interaction beyond gravity; if it is not, another medium of the bulk must carry the flow. Either way a source that couples only to gravity is not enough once the universe expands. **Verdict:** exact on the static bulk, not exact in an expanding universe. Status: DERIVED (ST4, ST6).

[2] It is absent in the primordial era, whose bulk is the author's vacuum (FE-31). The author's vacuum, $a_4 = Hx_4$ with constant b_4 ($\mathsf{A}=1$, $\mathsf{U}=0$), solves the field equations of the bulk with nothing on the right (ST2):

$$
\begin{aligned}
&a_4 = Hx_4, \quad b_4 = \text{constant } (\mathsf{A}=1,\ \mathsf{U}=\mathsf{A}_1=\mathsf{U}_1=0): \qquad \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^{M}{}_{N} + \Lambda\, \delta^{M}_{N} = 0 \quad (\sigma \neq 0),\\
&\kappa\, {\mathcal{S}^{(i)}}_{(i)} = \kappa\, {\mathcal{S}^{(r)}}_{(r)} = -10H, \qquad \kappa\, {\mathcal{S}^{(4)}}_{(4)} = -6H,\\
&\kappa\rho_{\text{bulk}} = -H^2\Big[\frac{3}{4}\big((\mathsf{A}-1-\mathsf{U})^2 + (\mathsf{A}-1)^2\big) + \dots\Big], \qquad \mathcal{G}(1,\mathsf{U}) = 2\mathsf{U}(3\mathsf{U}-2)
\end{aligned}
\tag{FE-31}
$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^{a}}_{b}$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-

dimensional normalisation, frame components along the fold); it is fixed by the jump of the first σ-derivatives of the metric across the fold (the junction conditions; $\kappa\mathcal{S}$ is the jump of the junction potential). Unit: GeV^7. Status: DERIVED.

- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; in a frame component such as ${\mathcal{S}^{(i)}}_{(i)}$ a direction x_i of ordinary space (the three components are equal); a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The second line is the source of the fold of this vacuum. The third gives the energy density of the bulk source that a metric close to the vacuum would need, to second order in the distances $\mathsf{A}-1$ and U from the vacuum (the dots stand for the terms of third and higher order; the exact expression, for any $a_4(x_4)$ and $b_4(x_4)$, is $\kappa\rho_{\mathrm{bulk}} = -H^2\,\mathsf{P}(\mathsf{A},\mathsf{U})/16$ with the polynomial P of FE-66 below), and, at $\mathsf{A} = 1$, the factor $\mathcal{G}$ of the flow of energy along the hidden direction (FE-66). In the primordial era no stabiliser is needed, and with the stabiliser of IN-7 present the author's vacuum would not be a solution. But SB-11 shows that a stabiliser that couples only to gravity and keeps $p = 5\rho$ cannot appear later out of nothing. **Verdict:** [2] is true. Together with [1] it requires a mechanism that creates the stabiliser between the primordial era and the late universe: either the exchange of energy with the fold of SB-12 (which breaks [1]), or a change of its equation of state (which a single medium with $p = 5\rho$ does not have). UM-3 contains no such mechanism. Status: DERIVED (ST2, ST6), OPEN (the mechanism).

[3] Its existence and value are DERIVED from the requirement of a static bulk. The value is DERIVED: SB-8 fixes every entry (ST1). The existence of a bulk source is DERIVED as well, and the check proves more than the requirement of an exactly static bulk needs: without any bulk source, the energy constraint of FE-66 allows no instant at which both rates are small (ST8). What is ASSUMED is that the source is a separate medium that couples only to gravity, and [1] shows where that assumption stops being exact. Its microscopic nature is OPEN; the only simple field that reproduces it is a ghost (SB-10, ST5). **Verdict:** value DERIVED; the existence of a bulk source DERIVED; its identification with a separate medium ASSUMED; its nature OPEN. (The record of the field equations of UM-3 of 2026-10-24 labelled the existence of the stabiliser ASSUMED and its value DERIVED; ST8 upgrades the existence of a bulk source to DERIVED.)

[4] The static bulk is defined in SB-5 and illustrated by five examples and a comparison of static and empty spacetimes in the part "The static bulk".

[5] Warp and warp factor are defined in SB-2 and illustrated by four examples in the part "Warp and warp factor".

8 The importance of the stabiliser for UM-3

Without a bulk source there is no late universe like ours. Without any bulk source, the author's metric with constant rates solves the bulk equations only at

$$\big(\mathsf{A}, \mathsf{U}\big) = (1, 0) \quad \text{or} \quad (-1, 0) \tag{SB-15}$$

Symbols in (SB-15).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.

(ST3): ordinary space must grow or shrink at the full rate H. Rates that change in time do not help. Two components of the bulk equations contain no second time derivative, for any $a_4(x_4)$ and $b_4(x_4)$ (they are constraints, like the Friedmann equation of cosmology): the time component, whose polynomial P fixes the energy density of the bulk source, and the flow of energy along the hidden direction (FE-66 of UM-3):

$$\kappa\,{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)} = \frac{H^2}{16}\,\mathsf{P}(\mathsf{A},\mathsf{U}), \qquad \kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(4)} = \frac{3}{16}\,\mathsf{U}\,\mathcal{G}(\mathsf{A},\mathsf{U})\,H^2\,\mathrm{sgn}(\sigma),$$
$$\mathsf{P} = 5\mathsf{A}^6 - 15\mathsf{A}^5\mathsf{U} + 15\mathsf{A}^4\mathsf{U}^2 - 9\mathsf{A}^4 - 5\mathsf{A}^3\mathsf{U}^3 + 18\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2 - 9\mathsf{A}\mathsf{U}^3 - 3\mathsf{A}\mathsf{U} - 3\mathsf{U}^2 + 1, \tag{FE-66}$$
$$\mathsf{P}(\mathsf{A},\mathsf{U}) = 0 \;\Rightarrow\; \mathsf{A}^2 + \mathsf{U}^2 \geq 0.2607, \qquad \mathsf{U} = \mathsf{A}: \quad 3a_4'^2 = 16\kappa\big(\rho_{\rm bulk} - \rho_{\rm s}\big)$$

Symbols in (FE-66).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\rm stab}$; ${\mathcal{T}_{\rm bulk}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathsf{P}(\mathsf{A},\mathsf{U})$: The polynomial of the energy constraint of the bulk: the (x_4, x_4) component of the bulk equations of the author's metric is $\kappa{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)} = (H^2/16)\,\mathsf{P}(\mathsf{A},\mathsf{U})$, whatever the time derivatives of the rates. Unit: dimensionless. Value: $\mathsf{P}(0,0) = 1$, $\mathsf{P}(\pm1, 0) = 0$. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum, 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\rho_{\rm bulk}$: The energy density of the bulk sources, $\rho_{\rm bulk} = -{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\rho_{\rm s}$: The energy density of the stabiliser; with frozen extra times it is the energy density of the bulk at which ordinary space stands still. Unit: GeV^8. Value: $\kappa\rho_{\rm s} = -H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).

Without a source $\mathsf{P} = 0$ at every instant. The smallest value of $\mathsf{A}^2 + \mathsf{U}^2$ on $\mathsf{P} = 0$ is 0.2607, reached at $(\mathsf{A},\mathsf{U}) = \pm(0.1432, 0.4901)$, and the smallest value of the larger of $|\mathsf{A}|$ and $|\mathsf{U}|$ on $\mathsf{P} = 0$ is 0.4771, reached at $\pm(0.2385, 0.4771)$: at no instant can both rates of a sourceless bulk be below $0.477H$ (ST8). A late universe that expands slowly (A small) and keeps its couplings constant ($\mathsf{U} = \mathsf{A}$) needs a bulk source; the static block needs exactly the stabiliser. ($\mathcal{G}(\mathsf{A},\mathsf{U}) = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$ is the factor of the flow; with frozen extra times the flow is the entry of FE-33.) Status: DERIVED (ST3, ST8).

The stabiliser is the zero point of the expansion. With frozen extra times ($\mathsf{U} = \mathsf{A}$) the polynomial is $\mathsf{P} = 1 - 3\mathsf{A}^2$, for any $a_4(x_4)$, and the energy constraint becomes a Friedmann equation of the bulk, the last formula of FE-66: $3a_4'^2 = H^2 + 16\kappa\rho_{\rm bulk} = 16\kappa(\rho_{\rm bulk} - \rho_{\rm s})$. The energy density of the stabiliser is the energy density of the bulk at which ordinary space stands still, and the rate at which ordinary space grows measures how far the energy density of the bulk lies above it. A bulk without any source cannot have frozen extra times at all: with $\mathsf{U} = \mathsf{A}$ the flow of FE-66 is $\frac{3}{16}\mathsf{A}(1+\mathsf{A}^2)H^2\,\mathrm{sgn}(\sigma)$, which vanishes only at $\mathsf{A} = 0$, where $\mathsf{P} = 1$ instead of 0 (a bulk source of zero energy density would need $a_4' = H/\sqrt{3}$, and it would still have to carry this flow). At constant A the pressures of the required source differ from those of the stabiliser only at the order A^2, and then differ between σ, ordinary space and the extra times (ST8). Status: DERIVED (ST8).

The alternative fails. One could try to make the author's vacuum itself the late universe by setting H equal to today's expansion rate $H_4^{(0)}$. But in the author's vacuum the extra times shrink, as $\mathrm{e}^{-a_4}b_4 = \mathrm{e}^{-Hx_4}b_4$, and the couplings of the Standard Model, whose fields are spread over the extra times, grow as their volume shrinks: every squared coupling grows as e^{3Hx_4}. The natural nuclear reactor of Oklo (Gabon), which ran about 1.8 billion years ago, bounds the change of the fine-structure constant $\alpha_{\rm em}$ since then to about one part in ten million. The author's vacuum would change it by

$$\Delta \ln \alpha_{\mathrm{em}} = 3H_4^{(0)}\, \Delta x_4 = 0.37 \tag{SB-16}$$

Symbols in (SB-16).

- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- α_{em}: The fine-structure constant. Unit: dimensionless. Value: 1/137.036. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

($H_4^{(0)} = 67.4$ km/s/Mpc, $\Delta x_4 = 1.8$ Gyr): more than a million times the bound. With the stabiliser and frozen extra times the couplings stay constant. Status: DERIVED (ST7), STANDARD (the bound of Oklo).

It powers the fold in an expanding universe. The same static block that requires the stabiliser in the bulk requires the thin source $\kappa\mathcal{S} = (-61H/8, -31H/4, -61H/8)$ on the fold (FE-32). When the universe expands, the energy that the bulk source releases is exactly the energy that the fold needs to keep its source (SB-13, FE-65, SB-14). Status: DERIVED (ST6).

It underlies every late-time result of UM-3. The late-time background of UM-3, the expansion of the metric around it, the bound states of dirac16complex on the fold and the cosmology on the fold are all computed on the static block or as small departures from it, and the static block is a solution only with the stabiliser. Status: DERIVED (the static block requires the stabiliser, ST1), ASSUMED (the static block as the late universe).

Its share of the energy. Integrated over the hidden direction and the extra times, per unit volume of ordinary space on the fold, the stabiliser carries the energy

$$\rho_{4,\mathrm{s}} = \frac{\rho_{\mathrm{s}} V_*}{3H} = -\frac{HV_*}{48\kappa} \tag{SB-17}$$

Symbols in (SB-17).

- $\rho_{4,\mathrm{s}}$: The energy of the stabiliser per unit volume of ordinary space of the fold: its energy density integrated over the hidden direction and the extra times. Unit: GeV^4. Value: $-HV_*/(48\kappa)$. Status: DERIVED.
- ρ_{s}: The energy density of the stabiliser; with frozen extra times it is the energy density of the bulk at which ordinary space stands still. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{s}} = -H^2/16$. Status: DERIVED (the value), ASSUMED (a separate medium).
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4} b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.

(the volume factor $\mathrm{e}^{-6H|\sigma|}$ of the warped directions, integrated over both sides of the fold, gives $1/(3H)$, and V_* is the physical volume of the extra times on the fold in the late universe). This is not a vacuum energy that makes the fold expand: the static block as a whole (the stabiliser, the curvature of the warped bulk, the cosmological constant Λ and the source of the fold together) does not make the fold expand at all. The observed slow expansion is a small departure from this balance, measured by the Friedmann equation of the bulk in FE-66. Status: DERIVED (ST7, ST8).

It is the most exotic ingredient of UM-3. Its negative energy density and the violation of the null energy condition (SB-9) put it among the sources that come with instabilities or with signals faster than light; Einstein's static universe, the example most like it, is unstable. If the stabiliser is a ghost field, its quanta carry negative energy, and even through gravity alone a ghost lets the vacuum decay into pairs of ghosts and pairs of ordinary particles such as photons, at a rate that is finite only if the momenta of the ghosts are cut off (Cline, Jeon and Moore, 2004). Whether the static block with the stabiliser is stable against small disturbances of the metric and of

the stabiliser itself has not been analysed. Status: STANDARD (the general results), OPEN (the stability of the static block with the stabiliser).

Summary. The stabiliser is the price of a slowly expanding late universe with constant couplings in the author's metric with the author's weights and cosmological constant. Its value is fixed exactly and introduces no new parameter; it is the zero point of the expansion of ordinary space; in an expanding universe it feeds the fold. Its nature, its origin between the primordial era and today, and its stability are the open questions that it leaves.

9 Provenance

What comes from the author (AUTHOR). The coordinates $X = \{x_0, \ldots, x_7\}$ and their roles (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, section DEFCHART); the metric with $b_4[x_4]^2$ and its two functions $a_4(x_4)$ and $b_4(x_4)$ (the author's order of 2026-10-24); the Lovelock weights $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ and the cosmological constant $\Lambda = -10H^2$ (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134], where the author writes the pure number 10); the author's vacuum $a_4 = Hx_4$ with constant b_4; the field dirac16complex.

What UM-3 adds. The proper distance σ (DERIVED from the metric); the static block as the first approximation of the late universe (ASSUMED); the stabiliser IN-7 (found and named by Claude on 2026-10-24; its value DERIVED); the fold fluid (ASSUMED, in the field equations of UM-3).

Who found what, and when. On 2026-10-24 Claude computed, in the gravity module of UM-3 (`um3_geometry.py`, check G4, commit 4448938 of 2026-10-24 17:54 UTC-7), that a static bulk of the author's metric needs the source SB-8, that the bulk with frozen extra times needs the energy density $(3\mathsf{A}^2 - 1)H^2/(16\kappa)$ (the content of FE-33 and of the last formula of FE-66) and the flow of energy SB-12 along the hidden direction. It introduced the medium and named it "the stabiliser of the bulk", as the term IN-7 of the action of UM-3, in the commit 170d8d3 of 2026-10-24 19:05 UTC-7 (the field equations of UM-3). On 2026-10-25, at the request of the author, who asked for a complete account of the stabiliser with proper attribution to its finder, Claude wrote this document with its LaTeX source and its PDF, wrote the checks ST1 to ST8, and found SB-10, SB-11, SB-13, SB-14, SB-15, the fold fluid FE-65 fed by the bulk, the constraints FE-66 for any $a_4(x_4)$ and $b_4(x_4)$ and their bound, and the reading of the Friedmann equation of the bulk as the zero point of the expansion. All of this work was done in Claude Code sessions configured for the model claude-opus-5-5 (session https://claude.ai/code/session_01AyG7BZ5rkLxBARRjLHHquB).

How it was computed. The left-hand side of the field equations was computed exactly with the computer algebra system sympy for the general diagonal warped metric $\mathrm{d}\sigma^2 + \mathrm{e}^{2\mathfrak{a}}(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2) - \mathrm{e}^{2\mathfrak{n}}\mathrm{d}x_4^2 - \mathrm{e}^{2\mathfrak{b}}(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2)$, with functions $\mathfrak{a}$, $\mathfrak{n}$, $\mathfrak{b}$ of σ and x_4: the curvature in an orthonormal frame, and the Lovelock tensors of orders 1, 2 and 3 contracted with the generalised Kronecker delta. It was then specialised to the author's metric, $\mathfrak{a} = a_4 - H|\sigma|$, $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$, $\mathfrak{n} = 0$. The source of the fold was computed from the jump of the first σ-derivatives of the metric across the fold (the junction conditions). The program `universe_model_3/code/um3_stabiliser.py` of the repository recomputes everything stated as DERIVED in this document; its eight checks are:

check	what it verifies	result
ST1	the static bulk requires the source SB-8 = IN-7, the same on both sides, without flow	PASS
ST2	the author's vacuum needs no source	PASS
ST3	without a source, constant rates are possible only at $(\mathsf{A}, \mathsf{U}) = (\pm 1, 0)$ (SB-15)	PASS
ST4	the stabiliser is conserved by itself on the static bulk	PASS
ST5	it violates the weak and null energy conditions (SB-9); only a ghost field reproduces it (SB-10)	PASS
ST6	SB-11; the flow SB-12; the energy budget SB-13; the fold fluid FE-65 and the balance of the fold SB-14	PASS

check	what it verifies	result
ST7	the drift SB-16 without it, and its share SB-17 of the energy	PASS
ST8	the constraints FE-66 contain no second time derivative; the bound of FE-66 and the Friedmann equation of the bulk	PASS

The program writes `outputs/stabiliser.json` and `reports/stabiliser_checks.json`, and runs in about 40 seconds (`python universe_model_3/code/um3_stabiliser.py` from the root of the repository). For SB-14 it reads the general source of the fold, which the fold program of UM-3 (`um3_fold.py`, check J5) computes exactly, from `outputs/fold.json`.

Sources outside UM-3 (STANDARD). A. Einstein, "Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie", Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (1917) 142 (the static universe, SB-6); A. S. Eddington, "On the instability of Einstein's spherical world", Monthly Notices of the Royal Astronomical Society 90 (1930) 668; K. Schwarzschild, "Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie", Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (1916) 189 (the static field outside a mass); D. Lovelock, "The Einstein tensor and its generalizations", Journal of Mathematical Physics 12 (1971) 498; R. V. Pound and G. A. Rebka, "Apparent weight of photons", Physical Review Letters 4 (1960) 337; L. Randall and R. Sundrum, Physical Review Letters 83 (1999) 3370 and 4690 (FD-8); S. Dubovsky, T. Grégoire, A. Nicolis and R. Rattazzi, "Null energy condition and superluminal propagation", Journal of High Energy Physics 03 (2006) 025; J. M. Cline, S. Jeon and G. D. Moore, "The phantom menaced: constraints on low-energy effective ghosts", Physical Review D 70 (2004) 043543; the bound of Oklo on the drift of the fine-structure constant, $-0.9 \times 10^{-7} < \Delta\alpha_{\mathrm{em}}/\alpha_{\mathrm{em}} < 1.2 \times 10^{-7}$ over 1.8 Gyr (A. I. Shlyakhter, Nature 264 (1976) 340; T. Damour and F. Dyson, Nuclear Physics B 480 (1996) 37); the Hubble constant $H_4^{(0)} = 67.4$ km/s/Mpc (Planck Collaboration, Astronomy and Astrophysics 641 (2020) A6).